MASTER STUDY LIBRARY INDEX

A Comprehensive Directory, Curriculum Sequence & Cross-Reference Roadmap for Technical Interviews

A Concise, Visual, and Mechanism-Focused Interview & Revision Handbook

METADATA FIELD	SPECIFICATION
Subject:	Technical Interview Preparation Library
Target Audience:	SDE / Technical Interviews (Top Tech, FAANG, Product Firms)
Methodology:	Internal Mechanics • State Transitions • Verified Code Recipes • Trade-offs
Library Scope:	Complete Personal Computer Science & Technical Interview Revision Library
Documents Consolidated:	7 Comprehensive Topic Handbooks + Master Audit Manifests
Key Function:	Topical Navigation, Cross-Reference Matrix, Study Tracks & Emergency 24h Review Checklist
Status:	Fully Generated, Programmatically Validated & Publication-Ready

1. Library Architecture & Document Catalog

Handbook Deliverable	Pages	Core Subject	Primary Interview Focus
DSA_Final_Master_Revision_Handbook.pdf	25	Data Structures & Algorithms	20 Chapters: Complexity, Memory Model, Two Pointers, Sliding Window, Monotonic Stacks, Fast/Slow, Binary Search, Heaps, Trees, Graphs, Backtracking, DP, Greedy, Bitwise, Tries, Decision Guide & Boilerplates.
Operating_Systems_Linux.pdf	12	Operating Systems & Linux	11 Chapters: Dual-mode, CPU scheduling, Synchronization (Mutex/Semaphore), Deadlocks (Banker's), Paging & TLB, epoll, Inodes.
Computer_Networks.pdf	11	Computer Networks	10 Chapters: OSI 7 Layers, TCP/IP, CIDR subnetting, Routing, TCP 3-way handshake, Flow & Congestion Control, TLS 1.3, WebSockets.
DBMS_SQL.pdf	9	DBMS & SQL	8 Chapters: Relational keys, Normalization 1NF-BCNF, B+ Tree indexing, ACID, Isolation anomalies, 2PL vs MVCC, SQL Window functions.
OOP_Concepts.pdf	10	Object-Oriented Programming	9 Chapters: 4 Pillars, Dynamic Dispatch, Duck typing, Multiple inheritance Diamond MRO, Abstract classes vs Interfaces, SOLID.
Redis_and_Caching.pdf	9	In-Memory & Distributed Caching	8 Chapters: In-memory architecture, RDB vs AOF, Data structures, Cache-Aside, Eviction (LRU/LFU), Distributed locks, Cluster slots.
System_Design.pdf	11	System Design & Distributed Architecture	10 Chapters: HLD vs LLD, Horizontal scaling, L4/L7 Load Balancers, API Gateway, Message Queues (Kafka), Rate Limiting, CAP/PACELC.
Machine_Learning_Revision.pdf	9	Machine Learning Foundations	8 Chapters: Supervised/Unsupervised, Model failure modes, Confusion Matrix formulas, Decision Trees, Random Forest, XGBoost, Pandas.
Python_Advanced_and_Core_Concepts.pdf	9	Python Internals & Engineering	8 Chapters: Decorators, `zip()`/`lambda`, LEGB Scopes, Mutability/Hashability, Python vs Java collections, Video/Image standards.

CHAPTER 2 - RECOMMENDED PREPARATION TRACKS & STUDY SEQUENCES

To maximize retention and performance during recruitment cycles, follow these structured study tracks based on remaining preparation time.

2.1 Track A: The 4-Week Intensive Foundation (Filter Rounds)

Phase	Timeline	Target Handbooks	Daily Mastery Goals
Week 1: Core Coding	Days 1 - 7	DSA Handbook (Ch 1 - 8) + Python Advanced	Two Pointers, Sliding Window, Prefix Sum, Monotonic Stacks, and Python list/dict complexity traps.
Week 2: Advanced DSA	Days 8 - 14	DSA Handbook (Ch 9 - 18)	Binary search on answer, Tree traversals, Heap dual-median, Backtracking archetypes, and 1D DP.
Week 3: Core CS Systems	Days 15 - 21	OS & Linux + Computer Networks	Process vs Thread, CPU scheduling, Mutex vs Semaphore, Paging/TLB, TCP 3-way handshake, and congestion control.
Week 4: Data & Architecture	Days 22 - 28	DBMS & SQL + OOP + Redis + System Design	Normalization (1NF-BCNF), B+ Trees, ACID & MVCC, SOLID principles, Caching patterns, and API Gateway scaling.

2.2 Track B: The Emergency 24-Hour Pre-Interview Refresh

When you have exactly 1 day before an on-site or virtual interview, review in this exact priority sequence:

1. **DSA Patterns Cheat-Sheet (DSA Handbook Chapter 18):** Re-read the Big-O complexity table and pattern recognition triggers.
2. **Operating Systems Core Invariants (OS Handbook Chapter 5, 6, 7):** Mutex vs Semaphore, 4 Coffman conditions, and TLB Effective Access Time math.
3. **Networks Lifecycle (CN Handbook Chapter 5, 6, 9):** TCP 3-way handshake sequence numbers, AIMD congestion phases, and TLS 1.3 handshake.
4. **DBMS Isolation Anomalies (DBMS Handbook Chapter 4 & 5):** Dirty Read, Non-Repeatable Read, Phantom Read vs Isolation Levels.
5. **Caching & System Design Blueprints (Redis Ch 5 & System Design Ch 5, 9):** Cache-Aside delete rule, Token Bucket rate limiting, and CAP Theorem.

CHAPTER 3 - CROSS-REFERENCE TOPICAL INDEX

Key engineering concepts frequently span multiple domains. Use this cross-reference index to examine how concepts connect across the curriculum.

Cross-Cutting Concept	Primary Handbook Location	Secondary / Cross-Domain Manifestations
LRU (Least Recently Used)	Operating_Systems_Linux.pdf (Ch 8)	• **DSA Handbook (Ch 10):** Implemented via Doubly Linked List + Hash Map. • **Redis_and_Caching.pdf (Ch 4):** Memory eviction policy (`allkeys-lru`).
Hashing & Hash Maps	DSA_Final_Master_Revision_Handbook.pdf (Ch 6)	• **DBMS_SQL.pdf (Ch 3):** Hash Indexing vs B+ Trees. • **Python_Advanced.pdf (Ch 4):** Open addressing, immutable keys. • **System_Design.pdf (Ch 7):** Consistent Hashing ring.
Concurrency & Synchronization	Operating_Systems_Linux.pdf (Ch 5)	• **DBMS_SQL.pdf (Ch 6):** Two-Phase Locking (2PL) vs MVCC. • **Redis_and_Caching.pdf (Ch 6):** Distributed Locks (SETNX PX + Lua script).
Rate Limiting & Throttling	System_Design.pdf (Ch 9)	• **Computer_Networks.pdf (Ch 6):** TCP Flow Control (rwnd) & Congestion Control (cwnd). • **Redis_and_Caching.pdf (Ch 3):** Sliding Window Log with Sorted Sets (ZSET).
Decoupling & Message Buffering	System_Design.pdf (Ch 8)	• **DSA Handbook (Ch 8):** FIFO Queue mechanics and bounded deques. • **Operating_Systems_Linux.pdf (Ch 9):** IPC message queues and Unix pipes.
Failure Handling & Timeouts	System_Design.pdf (Ch 10)	• **Computer_Networks.pdf (Ch 5):** TCP TIME_WAIT state and Retransmission Timer (RTO). • **Redis_and_Caching.pdf (Ch 7):** Cache stampede mutex locks and TTL jitter.

MASTER CURRICULUM VERIFICATION AUDIT

All 9 generated revision handbooks have been compiled strictly within `Daily-Progress/documents/` using ReportLab 4.5.1. Every code template has been syntax-checked and validated against boundary test cases. Original source materials remain completely unmodified and preserved.